

Viraat Mindaguditi

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Summary

Machine learning engineer with hands-on experience in extracting insights from large datasets and building data models. Skilled in Python, SQL, Power BI, and statistical analysis to develop predictive models and interactive dashboards. Improved weather prediction accuracy by 23% and automated reporting processes to support data-driven decisions.

Education

University of Houston

Master's degree, Computer Science

Aug 2024 - May 2026

Indian Institute of Information Technology, Sricity

Bachelor's degree, Computer Science and Engineering

Nov 2020 - Jul 2024

Experience

Law Information Technology University of Houston *Machine Learning Engineer - IT Automation* **May 2025 - Present**

- Designed and deployed Power BI dashboards to visualize system health metrics, ticket trends, and performance KPIs, enabling data-driven infrastructure decisions through advanced data visualization.
- Resolved over 200 technical support tickets involving troubleshooting, network connectivity, and user access control by applying data analytics to identify common issues and improve resolution workflows.
- Automated recurring administrative tasks using PowerShell and Python, reducing manual effort by 30%.
- Provided Tier-1 and Tier-2 support to 700+ students and faculty, achieving a 95%+ customer satisfaction rate.

Cullen College of Engineering University of Houston | *Research Assistant*

Oct 2024 - Apr 2025

- Contributed to the development of a scalable web application for the Illinois Department of Transportation using Angular, improving user interaction and responsiveness across modules through statistical analysis of user engagement data.
- Implemented key front-end features for calculating and displaying equipment specifications, integrating dynamic forms and reusable components with TypeScript and RxJS.
- Designed and maintained MongoDB data models and collections for equipment data and pricing to optimize data retrieval.
- Engineered RESTful API integrations to streamline client-server communication and enhance system modularity.

ISRO (Indian Space Research Organisation) | *Data Science Intern*

Aug 2023 - Apr 2024

- Increased the accuracy of weather prediction project by 23% using deep-learning techniques using real-time data for sounding rocket launches.
- Stored multiple factors of predicted data into the database for easy assessment through graphs.
- Developed and installed a complaint page on the SHAR intranet website, simplifying the user feedback processes.

Key Projects

Drug Target Interaction Prediction using GCNNs

Apr 2023 - Nov 2023

- Developed a cutting-edge drug-target interaction model using graph convolutional neural networks (GCNNs), which processed over 17000 DTI pairs from biosnap database.
- Implemented Web scraping technique to extract drug smiles and protein structures from specific websites.
- Innovated a preprocessing system to graphically represent 10,000+ protein sequences and drug SMILES that helped in the rapid development of GCNN models.
- Used various libraries like torch, pandas, sklearn, networkx, numpy, rdkit for web scraping and preprocessing.

StartCom Social Networking Site

Jul 2022 - Dec 2022

- Designed and implemented an innovative platform enabling users to publish posts and engage in live chats.
- Implemented login/signup page by linking with reducers with appropriate validations.
- Used ReactJS, Redux, and socket programming to implement various features of the website.
- Configured an efficient MongoDB server to maintain user registration details, enhancing access control through specific admin credentials.

Airport Management System

Aug 2021 - Dec 2021

- Created a command line application using Java to manage the database of pilots, flights, passengers, and hotels in airports.
- Used MySQL database management system to store and display the details.
- Implemented CRUD operations using object-oriented programming concepts in Java.
- Developed the application that links the details of pilots to flights and passengers to hotels.

Technical Skills

- Programming Languages:** C/C++, Python, Java, HTML/CSS, JavaScript, MATLAB, SQL
- Tools & Frameworks:** Power BI, Git, ReactJS, Redux, NodeJS, Simulink, MySQL, MongoDB
- Libraries & Packages:** NumPy, pandas, Matplotlib, Sklearn, Networkx, torch, tensorflow, rdkit
- Data Analysis & Modeling:** Data Analytics, Data Modeling, Statistics, Data Visualization
- Industry Experience:** Consulting Experience, Tech Industry Experience